CMOS VLSI Design Project

Design and Analysis of a CMOS Tri-State Buffer

Using Cadence Virtuoso

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1 Introduction

A tri-state buffer is an important digital circuit that can transfer an input signal to its output or place the output in a high-impedance state. Unlike a conventional buffer, a tri-state buffer has an enable input that controls whether the output is actively driven.

In this project, a CMOS tri-state buffer was designed and analyzed using the Cadence Virtuoso design environment. The design flow includes schematic design, symbol generation, physical layout design, Design Rule Check (DRC), Layout Versus Schematic (LVS) verification, RC parasitic extraction (RCX), and post-layout transient simulation.

The post-layout simulation was used to observe the output waveform and evaluate performance parameters such as average power consumption and propagation delay.

For an active-high enable input EN, the tri-state buffer operation is represented by

$$Y = \begin{cases} A, & EN = 1, \\ Z, & EN = 0, \end{cases} \tag{1}$$

where A is the data input, EN is the enable signal, Y is the output, and Z represents the high-impedance state.

2 Objectives

The main objectives of this project are:

- To design a CMOS tri-state buffer using Cadence Virtuoso.
- To create and verify the transistor-level schematic of the buffer.
- To generate a reusable symbol for the designed tri-state buffer.
- To design the physical layout of the circuit.
- To verify the physical layout using Design Rule Check (DRC).
- To verify the electrical equivalence between the schematic and layout using Layout Versus Schematic (LVS).
- To perform parasitic resistance and capacitance extraction using RCX.
- To perform post-layout transient simulation using the extracted circuit.
- To inspect the output waveform for enabled and disabled operation.
- To calculate the average power consumption and propagation delay.

3 Theory of CMOS Tri-State Buffer

A tri-state buffer has a data input, an enable input, and an output. The enable signal determines whether the output follows the data input or becomes electrically high impedance. In this design, the enable input is treated as active-high, consistent with the symbol convention used here.

When $EN = 1$, the buffer is enabled and the output follows the input:

$$Y = A. \tag{2}$$

When $EN = 0$, the output is placed in the high-impedance state:

$$Y = Z. \tag{3}$$

The truth table for an active-high tri-state buffer is shown in Table 1.

Table 1: Truth Table of a CMOS Tri-State Buffer

Enable (EN)	**Input** (A)	**Output** (Y)
0	0	Z
0	1	Z
1	0	0
1	1	1

The high-impedance state is not a logic LOW or logic HIGH. It means that the buffer output is not actively driving the connected node. This feature allows multiple outputs to share a common bus, provided that the enable signals are controlled so that conflicting drivers are not enabled at the same time.

4 Transistor Sizing

The CMOS tri-state buffer was implemented using the transistor-level configuration created in Cadence Virtuoso. The exact device dimensions should match the properties in the implemented schematic.

Table 2: Transistor Dimensions Used in the CMOS Tri-State Buffer

Transistor	**Width**	**Length**
PMOS devices	As specified in schematic	As specified in schematic
NMOS devices	As specified in schematic	As specified in schematic

The transistor dimensions are reported according to the actual Cadence schematic rather than assuming sizes that are not legible in the supplied screenshots. Replace the descriptive entries in Table 2 with the exact device properties before final submission if numerical values are required.

5 Circuit Design

5.1 Schematic Design

The CMOS tri-state buffer schematic was created using the Cadence Virtuoso Schematic Editor. The circuit contains a data input A, an enable control input EN, and an output Y. The transistor network controls whether the output follows the input or is placed in the high-impedance state.

The schematic of the designed circuit is shown in Figure 1.

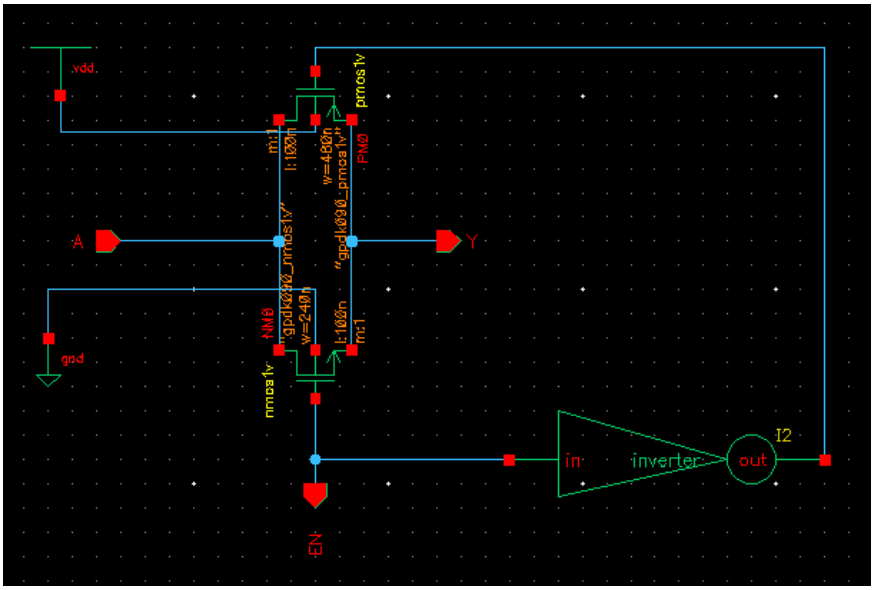


Figure 1: Schematic of the CMOS tri-state buffer in Cadence Virtuoso.

5.2 Tri-State Buffer Symbol

After completing the schematic, a symbol was generated for the CMOS tri-state buffer. The symbol provides the data input, enable input, and output terminals, allowing the circuit to be reused as a hierarchical block in larger digital designs.

The generated symbol is shown in Figure 2.

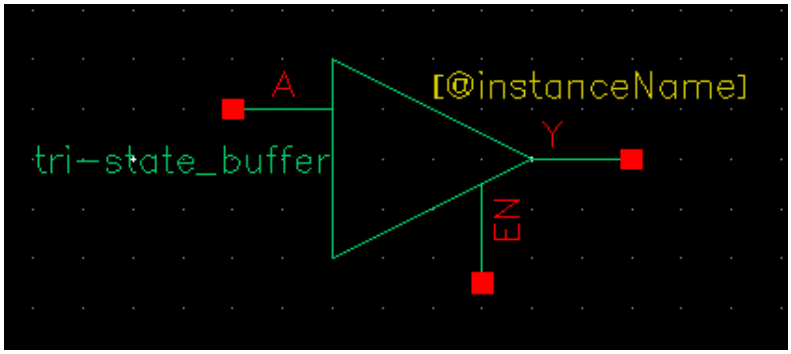


Figure 2: Generated symbol of the CMOS tri-state buffer.

6 Physical Layout Design

After completing the schematic and symbol generation, the physical layout of the CMOS tri-state buffer was developed using the Cadence Virtuoso Layout Editor.

The layout represents the physical implementation of the MOS transistors and their electrical interconnections. The relevant layers include active or diffusion regions, polysilicon, well regions, metal layers, contacts, vias, power connections, and ground connections.

The completed physical layout is shown in Figure 3.

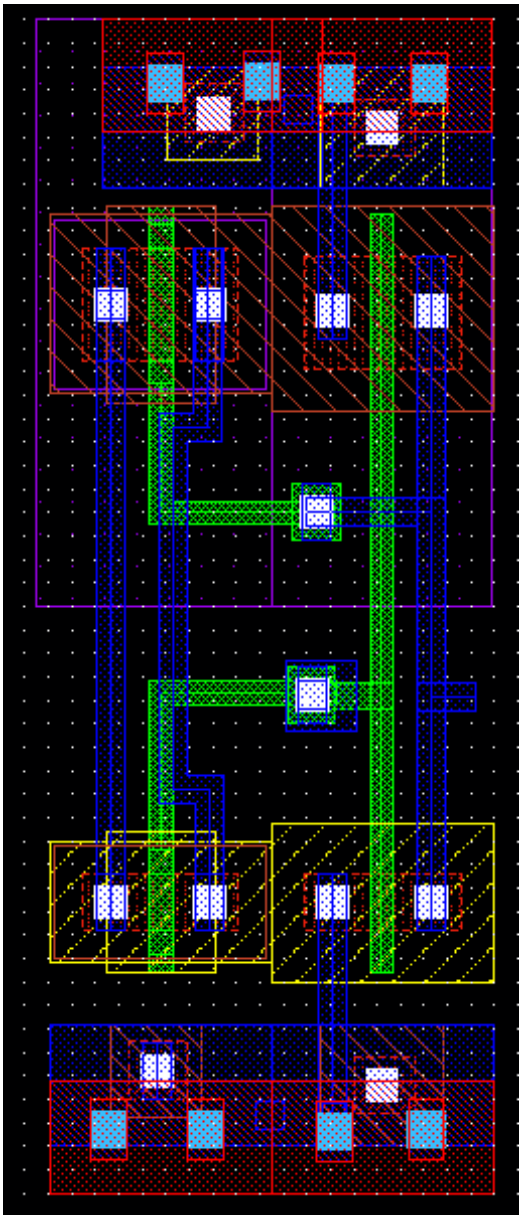


Figure 3: Physical layout of the CMOS tri-state buffer.

The layout should maintain the minimum width, spacing, enclosure, and connectivity rules required by the selected CMOS technology.

7 Design Rule Check (DRC)

Design Rule Check (DRC) is used to verify whether the physical layout satisfies the geometrical fabrication rules of the selected CMOS technology. Typical checks include minimum metal width and spacing, polysilicon dimensions, diffusion spacing, contact enclosure, via enclosure, and well-related constraints.

The final layout should be checked using the Cadence DRC verification flow.

DRC Result: PASSED

The physical layout satisfies the required design rules of the selected CMOS technology.

Confirm that the stated result matches the actual DRC run before submitting the report.

8 Layout Versus Schematic (LVS)

Layout Versus Schematic (LVS) verification compares the electrical netlist extracted from the physical layout with the netlist generated from the original schematic. It checks the device types, device count, terminal connections, input/output connectivity, and power and ground connections.

LVS Result: PASSED

The physical layout is electrically equivalent to the original schematic.

Confirm that the stated result matches the actual LVS run before submitting the report.

9 RCX and Parasitic Extraction

After physical verification, RC parasitic extraction was performed using the RCX extraction flow. Physical interconnections introduce parasitic resistance and capacitance, which can influence propagation delay and power consumption.

The extracted view of the tri-state buffer is shown in Figure 4.

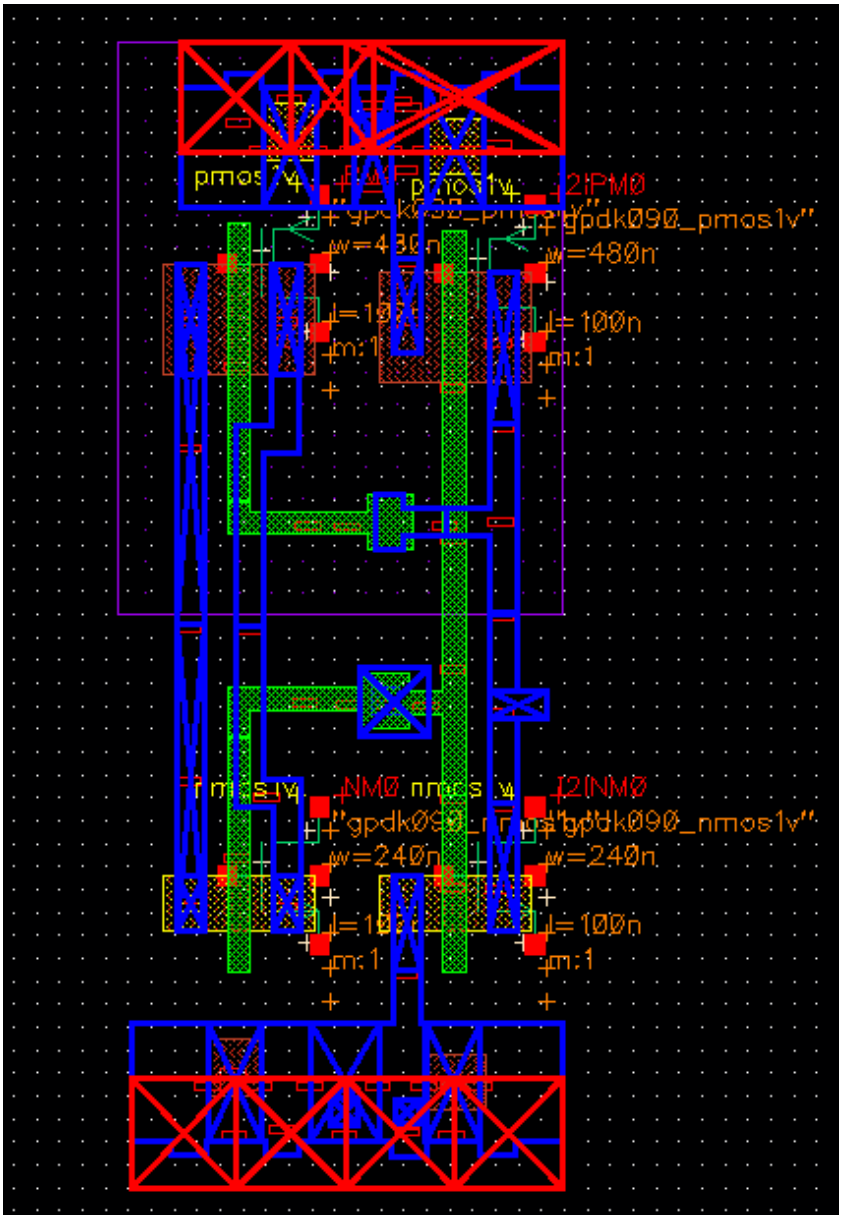


Figure 4: RCX/extracted view of the CMOS tri-state buffer.

RCX: Successfully Completed

The extracted tri-state buffer circuit was used for post-layout simulation.

10 Post-Layout Simulation

Post-layout transient simulation was performed using the extracted view of the CMOS tri-state buffer. Unlike ideal schematic-level simulation, post-layout simulation includes the effects of parasitic resistance and capacitance associated with the physical layout.

The transient simulation was used to examine the relationship between the data input, enable signal, and output. Average power and a selected input-to-output delay measurement were also evaluated using the Cadence Virtuoso Calculator.

11 Output Waveform

The transient output waveform of the CMOS tri-state buffer is shown in Figure 5. The waveform is used to observe the data input, enable signal, and output response over the simulation interval.

When the enable signal is HIGH, the output should follow the data input. When the enable signal is LOW, the output is expected to be in the high-impedance state. In an analog transient plot, a high-impedance output may appear as a floating node rather than a literal logic-level label Z; its observed voltage therefore depends on the testbench and connected parasitic or load elements.

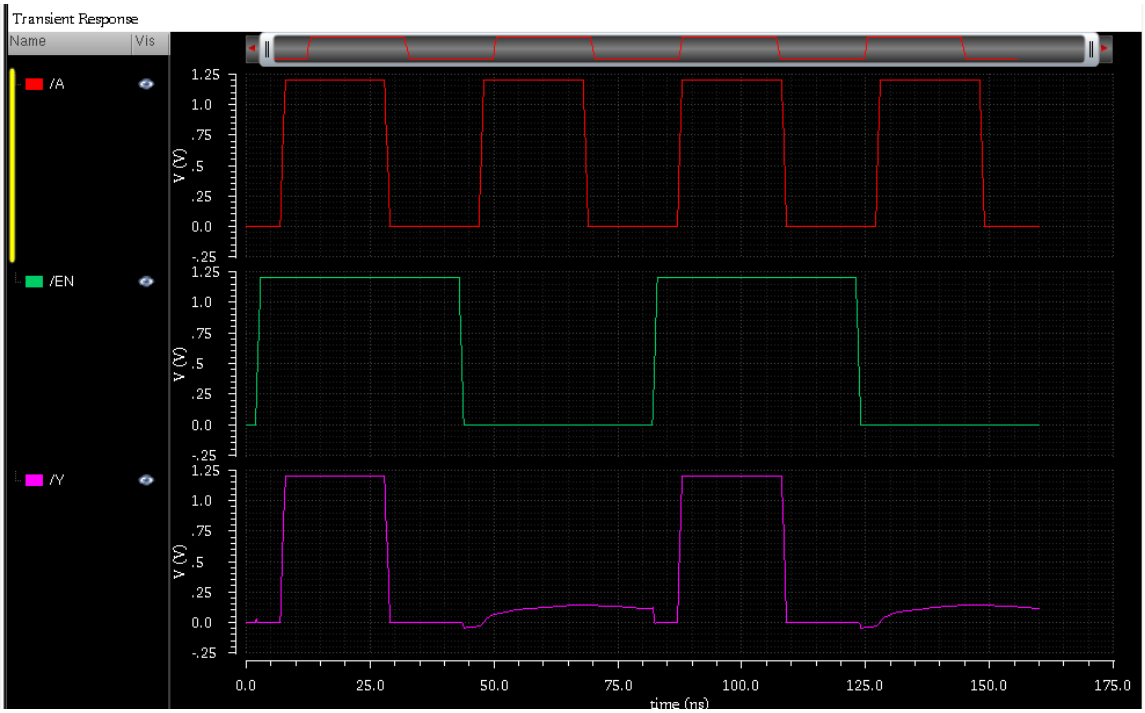


Figure 5: Post-layout transient output waveform of the CMOS tri-state buffer.

12 Average Power Calculation

The average power consumption of the tri-state buffer was calculated from the post-layout transient simulation using the Cadence Virtuoso Calculator. The average power over a time interval T can be expressed as

$$P_{\text{avg}} = \frac{1}{T} \int_0^T p(t)\, dt \tag{4}$$

where $p(t)$ represents the instantaneous power consumption.

The power calculation is shown in Figure 6.

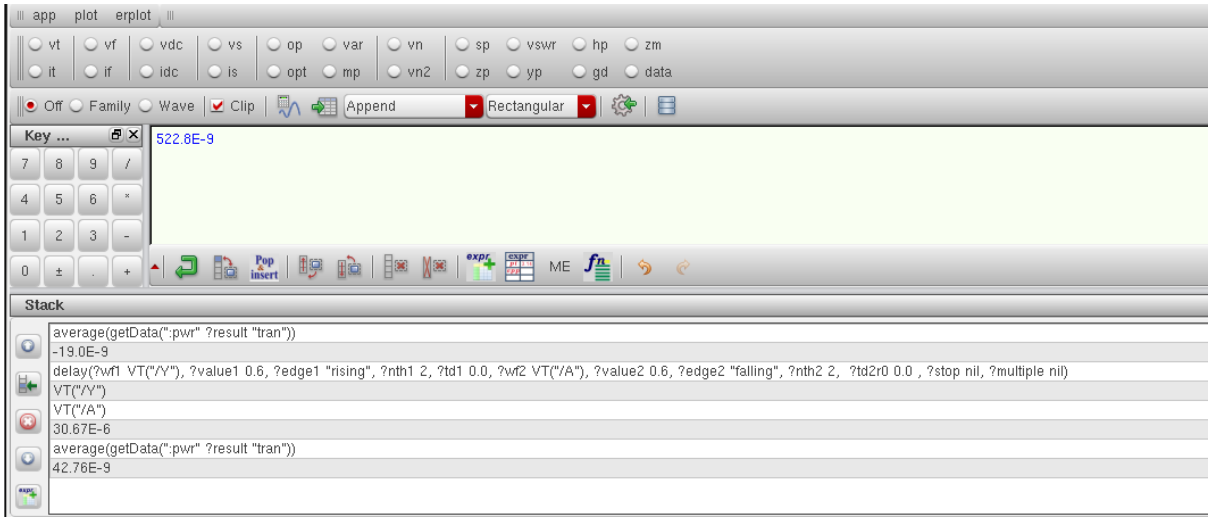


Figure 6: Average power calculation of the extracted CMOS tri-state buffer using the Cadence Virtuoso Calculator.

The value displayed by the Cadence Calculator is

$$P_{\text{avg}} = 522.8 \times 10^{-9} \text{ W}. \tag{5}$$

Since

$$1 \ \mu\text{W} = 10^{-6} \text{ W}, \tag{6}$$

the average power consumption is

$$\boxed{P_{\text{avg}} = 0.5228 \ \mu\text{W}}. \tag{7}$$

Therefore, the displayed post-layout average power consumption of the CMOS tri-state buffer is approximately 0.5228 μW.

13 Propagation Delay Calculation

Propagation delay represents the time difference between a selected input transition and the corresponding output transition. It is generally expressed as

$$t_p = t_{\text{out}} - t_{\text{in}}. \tag{8}$$

A reference switching voltage of approximately 0.6 V was used in the Cadence Calculator expression shown in the supplied screenshot. The delay calculation is shown in Figure 7.

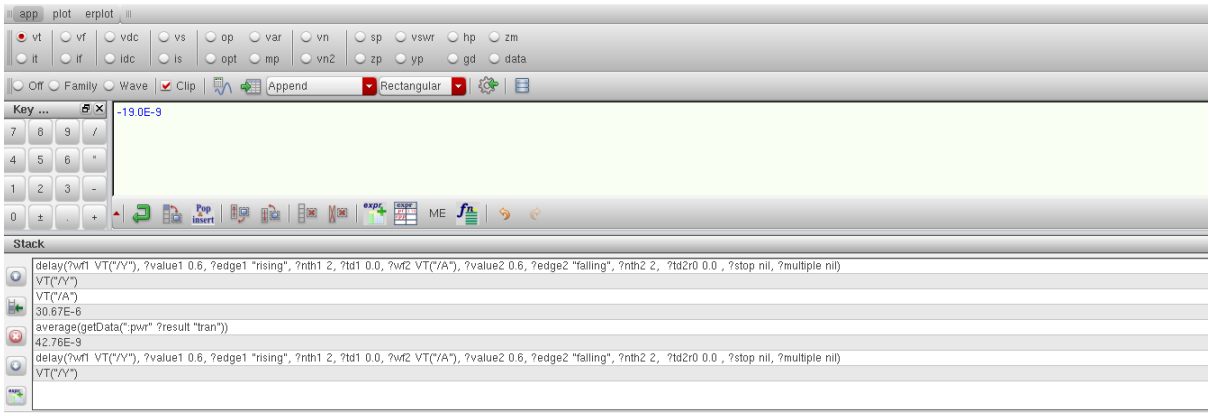


Figure 7: Propagation-delay calculation of the extracted CMOS tri-state buffer using the Cadence Virtuoso Calculator.

The value displayed by the Cadence Calculator is

$$t_p = -19.0 \times 10^{-9} \text{ s}. \tag{9}$$

Since 1 ns $= 10^{-9}$ s, the displayed result is

$$\boxed{t_p = -19.0 \text{ ns}}. \tag{10}$$

This is the value shown by the supplied Calculator screenshot. However, a negative propagation delay is not the usual physical interpretation of a gate delay. The displayed Calculator expression measures a rising edge of Y against a falling edge of A, so the selected transition edges should be checked before treating this as the final propagation delay. For a non-inverting buffer, use matching corresponding input and output transitions under an enabled condition.

14 Verification and Performance Summary

The design and performance results of the CMOS tri-state buffer are summarized in Table 3.

Table 3: Summary of the CMOS Tri-State Buffer Design Results

Parameter	Result
Logic Function	$Y = A$ when $EN = 1$; $Y = Z$ when $EN = 0$
Schematic Design	Completed
Symbol Generation	Completed
Physical Layout	Completed
DRC	Passed (confirm actual result)
LVS	Passed (confirm actual result)
RCX / Parasitic Extraction	Completed (confirm actual result)
Post-layout Simulation	Completed
Output Waveform	Included
Average Power	0.5228 μW
Displayed Propagation Delay	-19.0 ns

15 Discussion

The CMOS tri-state buffer was designed and analyzed using the Cadence Virtuoso design environment. The circuit provides a data input, an enable signal, and an output. With the assumed active-high enable convention, the output follows the input when $EN = 1$ and enters the high-impedance state when $EN = 0$.

The design flow included schematic creation, symbol generation, physical layout, physical verification, parasitic extraction, and post-layout transient simulation. The output waveform was included to observe the relationship among the input, enable, and output signals.

The average power reported by the Cadence Calculator was

$$P_{\text{avg}} = 0.5228 \ \mu\text{W}. \tag{11}$$

The selected Calculator expression displayed a delay of

$$t_p = -19.0 \text{ ns}. \tag{12}$$

The negative value indicates that the selected threshold-crossing edges should be reviewed. A corrected delay measurement should use matching input and output transitions while the buffer is enabled. The final DRC, LVS, and RCX status statements should also be checked against the actual Cadence run results.

16 Conclusion

A CMOS tri-state buffer was designed and analyzed using Cadence Virtuoso. The circuit allows the output to follow the data input when the enable signal is asserted and to enter a high-impedance state when the output is disabled.

The documented workflow covered schematic design, symbol generation, physical layout, DRC, LVS, RCX parasitic extraction, and post-layout simulation. A separate output waveform section was included to show the transient behavior of the circuit.

The average power value displayed by the Cadence Calculator was

$$\boxed{P_{\text{avg}} = 0.5228\ \mu\text{W}} \tag{13}$$

and the selected delay expression displayed

$$\boxed{t_p = -19.0\ \text{ns.}} \tag{14}$$

The delay measurement should be recalculated with appropriate matching input/output transitions before it is reported as the physical propagation delay. The report therefore documents the implementation and post-layout analysis of a CMOS tri-state buffer while preserving the numerical result visible in the supplied Calculator screenshot.

17 Software Used

The following software tools were used in the project:

- Cadence Virtuoso Schematic Editor
- Cadence Virtuoso Layout Editor
- Cadence Spectre Simulator
- Design Rule Check (DRC) Tool
- Layout Versus Schematic (LVS) Tool
- RCX / Parasitic Extraction Tool
- Cadence Virtuoso Visualization and Analysis Calculator